

Differences Between Virtualization and Cloud Computing

Objectives

After completing this reading, you will be able to:

- Describe virtualization and cloud computing
- Explain the difference between virtualization and cloud computing

Introduction

Virtualization is a process that enables more effective utilization of physical computer hardware and is the underlying basis of cloud computing. However, cloud computing can be defined as the on-demand access of computing resources, such as data storage, physical or virtual servers, networking capabilities, software, application development tools, and so on over the internet, available on pay-per-use pricing.

Learning about the distinctions between virtualization and cloud computing is crucial as it enables you to opt for the right technology to make informed decisions about your computing needs. Let's take a look at the key differences.

Feature	Virtualization	Cloud computing
Definition	Contains information about events that have occurred within a software application	A service that delivers computing resources (servers, storage, databases, networking, and software) over the internet
Dependency on hardware	Highly dependent on physical hardware	Minimal hardware dependency if using cloud services
Scope	Primarily focused on server or client virtualization	Offers various services, including storage, processing, and networking capabilities
Purpose	To maximize the use of computing resources by dividing a single server into multiple virtual machines	To provide scalable computing resources on demand
Access	Accessed through local networks or directly on the host machine. Does not require internet access for the obtained software	Accessed through the internet, allowing for mobility and remote usage
Customization and access control	Offers more control and customization over the virtualized environments	Less control over the infrastructure, as the service provider manages it
Use case	Used for consolidating physical servers, hosting multiple applications on a single machine, and running different operating systems for testing purposes	Ideal for businesses looking for scalable solutions, such as web hosting, extensive data analysis, and cloud-based applications

Summary

In this reading, you learned that:

- Virtualization creates multiple 'virtual' instances or environments from a single, physical hardware system. Its focus is on maximizing the efficiency of the existing physical resources.
- Whereas cloud computing is a service model that helps you with on-demand access to shared computing resources like applications, servers, and data storage over the internet, emphasizing ease of use, scalability, and reduced IT costs.



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